

Measuring and adjusting the weathering and hydraulic sorting effects for rigorous provenance analysis of sedimentary rocks: a case study from the Jurassic Ashikita Group, south-west Japan

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ABSTRACT

The present study examines the provenance of the Jurassic Ashikita Group distributed in south-west Japan, which is composed of the Idenohana, Kyodomari and Sakamoto Formations. Two geochemical diagrams for provenance analysis were utilized, which incorporate full consideration of compositional modifications resulting from weathering (MFW diagram) and hydraulic sorting processes ($\text{SiO}_2/\text{Al}_2\text{O}_3\text{--Na}_2\text{O}/\text{K}_2\text{O}$ diagram). The MFW diagram delineates weathering trends of sedimentary rocks and allows estimation of the original source rock composition by tracing the weathering trends backwards to an unweathered domain. Weathering trends of the Idenohana and Kyodomari Formations extend backward to the domain of intermediate and felsic igneous rocks. In contrast, sediments of the Sakamoto Formation do not fit into a linear weathering trend, indicating that the source rock cannot be approximated to igneous rocks. On the $\text{SiO}_2/\text{Al}_2\text{O}_3\text{--Na}_2\text{O}/\text{K}_2\text{O}$ diagram, sediments are organized into compositional trends, in which the range reflects compositional variations induced by the hydraulic sorting effect. On this diagram, sediments derived from the igneous and recycled sedimentary provenances can be distinguished by reading the inclination of the trend. By utilizing this principle, source rocks of the Idenohana and Kyodomari Formations are interpreted as igneous rocks and those of the Sakamoto Formation are interpreted as recycled sedimentary rocks. Therefore, these diagrams concurrently estimate the source rock composition through quantifying and adjusting the weathering and sorting effects, and reveal a systematic transition in the provenance of the Ashikita Group. The Idenohana and Kyodomari Formations were supplied chiefly from an igneous provenance, which shifted from intermediate to felsic compositions in stratigraphic order. Whereas, sediments of the Sakamoto Formation were sourced primarily from a recycled sedimentary provenance.

Keywords Ashikita Group, geochemistry, hydraulic sorting, Kurosegawa Belt, provenance, weathering.

INTRODUCTION

Chemical and modal compositions of sedimentary rocks provide important information for reconstructing palaeoclimate, tectonic setting and provenance of sedimentary basins (Dickinson & Suczek, 1979; Nesbitt & Young, 1982; Bhatia, 1983; Dickinson *et al.*, 1983; Roser & Korsch,

1988; Johnsson, 1993; Tabor *et al.*, 2004). The geochemical properties of sedimentary rocks have acquired particularly great interest, partly because the geochemical analysis technique is applicable to a wide range of grain-size populations, which is a definite advantage in avoiding biased sampling (Roser & Korsch, 1986; Nesbitt & Young, 1996; Ohta, 2004; Weltje & von Eynatten,

2004). For example, modal analysis using microscopes can only be applied to medium-grained to coarse-grained sandstones, but there is no guarantee that this grain-size population reflects an averaged composition of the bulk sedimentary unit. In addition, recent studies have demonstrated the discriminative potential and sensitivity of geochemical data against many factors, such as source-area lithology, weathering, diagenesis, hydraulic sorting and tectonic regime of the basin (e.g. Bhatia, 1983; Roser & Korsch, 1986; von Eynatten *et al.*, 2003b).

However, this sensitiveness in return means that the geochemical composition of sediments represents a complex outcome controlled by multiple causes. It is, therefore, difficult or even impossible to obtain any direct provenance information from raw geochemical data (e.g. Johnsson, 1993). It is especially necessary to measure and adjust the geochemical modifications induced by weathering and hydraulic sorting processes prior to any provenance interpretation (e.g. Ingersoll *et al.*, 1984; Argast & Donnelly, 1987; Garcia *et al.*, 1991; Nesbitt & Young, 1996; Nesbitt *et al.*, 1996; Ohta, 2004).

In this context, Nesbitt & Young (1984) and Fedo *et al.* (1995) described an innovative approach for provenance analysis that implicitly includes adjustment for the source-area weathering effect using the A-CN-K diagram, which predicts weathering trends of feldspars. These weathering trends can be used for a backward estimate of the unweathered source rock composition. Although the advantages of such backward-estimate methods have been stressed by many studies (e.g. von Eynatten, 2004), the A-CN-K diagram has been the only conventional diagram available for such investigative approaches.

Ohta & Arai (2007) described the MFW ternary diagram, which also predicts the weathering trends of igneous rocks. The M and F vertices of this ternary diagram characterize mafic and felsic igneous parent rocks, respectively, while the W vertex identifies the degree of weathering of these sources. Weathering trends depicted on the MFW diagram are mathematically independent of geochemical variability induced from the parent rock types (Ohta & Arai, 2007) and, thus, it may distinguish weathering trends of sediments sourced from a variety of igneous rock types.

The purpose of the present study is to examine the effects of source-area weathering and hydraulic sorting processes and to reveal provenance information, using as a case study the Jurassic Ashikita Group from south-west Japan. Source

rock lithologies of the sediments in the Ashikita Group can be inferred easily from the modal composition. Therefore, these sediments were suitable for investigating the manner in which weathering and sorting effects influence the source rock composition. The MFW diagram will be utilized to measure and adjust for the effect of source-area weathering. In addition, the $\text{SiO}_2/\text{Al}_2\text{O}_3\text{--Na}_2\text{O/K}_2\text{O}$ diagram (Ohta, 2004) will be used to discuss and correct for the effect of the hydraulic sorting process.

REGIONAL GEOLOGY

The Japanese Islands constituted a part of the eastern Asian continental margin before the opening of the Japan Sea in the Miocene (e.g. Otofuiji & Matsuda, 1987). The principal framework of the Japanese Islands is composed of late Palaeozoic to Cenozoic accretionary complexes and subduction-related igneous rocks that have been formed by the westward subduction of the palaeo-Pacific plates (e.g. Taira & Tashiro, 1987; Isozaki, 1996).

The study area is located in Kyushu Island, the westernmost main island of Japan, where the Kurosegawa Belt is distributed (Fig. 1). The Kurosegawa Belt is a disrupted belt, which is dis-

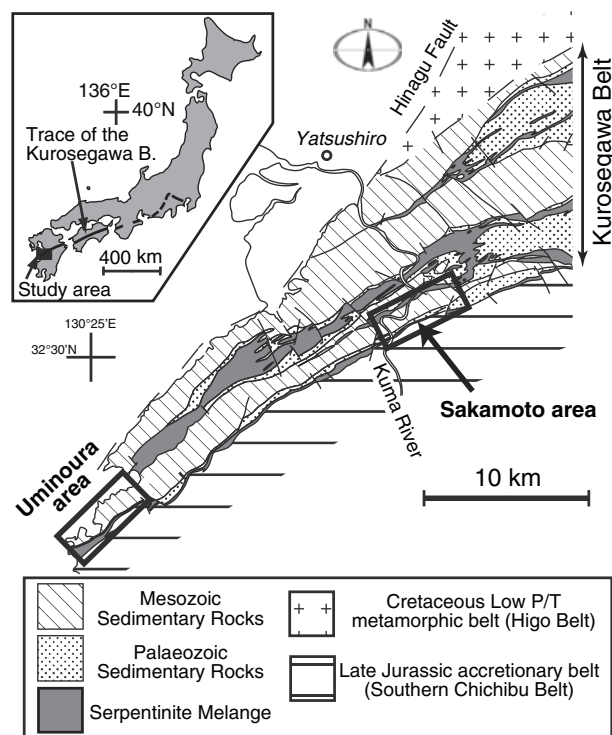


Fig. 1. Geological sketch map of the Kurosegawa Belt in western Kyushu (partly modified after Matsumoto & Kanmera, 1964).

tinctly different from other Palaeozoic to Cenozoic accretionary belts of SW Japan. A highly specific lithological assemblage is recognized in the Kurosegawa Belt, including Palaeozoic felsic to ultramafic igneous rocks, Palaeozoic high-*P/T* to low-*P/T* metamorphic rocks, Permian and Jurassic accretionary complexes and Silurian to Cretaceous sedimentary successions, which all generally occur as blocks within the serpentinite melange (Ichikawa *et al.*, 1956; Maruyama, 1981; Maruyama *et al.*, 1984). These rocks are distributed within a narrow belt (*ca* 10 km in width), but the belt intermittently extends for more than 1500 km along the Japanese Islands (Fig. 1). Although various models have been proposed to account for the occurrence of such a complicated belt (e.g. Maruyama *et al.*, 1984; Taira & Tashiro, 1987; Isozaki & Itaya, 1991; Sakai & Okada, 1997; Kato & Saka, 2003), the origin and evolution of the Kurosegawa Belt remain uncertain.

The provenance analysis of Mesozoic siliciclastic sediments of the Kurosegawa Belt is important because it may provide some insights into the hinterland geology of the belt. Such information will aid reconstruction of the palaeogeographical conditions of the Kurosegawa Belt. In the western part of Kyushu, fairly coherent Mesozoic sedimentary rocks are distributed widely, which is favourable for provenance analysis (Fig. 1). In the study areas (Uminoura and Sakamoto areas), the Jurassic Ashikita Group is composed of the Lower Jurassic Idenohana Formation, Middle Jurassic Kyodomari Formation and upper Middle Jurassic to Upper Jurassic Sakamoto Formation (Fig. 2, Ohta & Sakai, 2003). Each of the three formations exhibits marked fining-upward cycles; the coarse-grained sedimentary gravity flow deposits located in the lower part are capped by the fine-grained hemipelagic drape deposits (Ohta & Sakai, 2003).

The Idenohana Formation mainly consists of conglomerates and sandstones comprising a channel-levée complex (Fig. 2). Clasts in conglomerates are composed chiefly of andesite, biotite granite, quartz diorite, limestone and mudstone with subordinate amounts of hornblende porphyry and dolerite (Kanmera, 1961). The Kyodomari and Sakamoto Formations mainly consist of turbidites, which indicate preferential southward sediment transportation (Fig. 2).

MATERIALS AND METHODS

In this study, 27 sandstone and 27 mudstone samples were collected for XRF analysis using

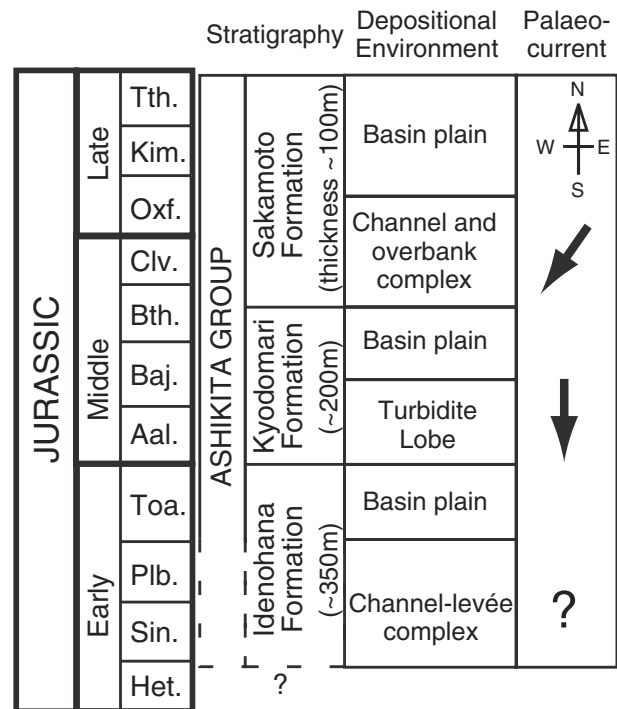


Fig. 2. Stratigraphy of the Jurassic Ashikita Group with depositional environments and palaeocurrent directions (Ohta & Sakai, 2003). North is up for palaeocurrents.

RIX2100 (Rigaku Co., Tokyo, Japan) at the Kurashiki University of Science and the Arts in Okayama Prefecture, Japan. Major and trace element data were obtained by fused and pressed beads, respectively. The mudstones collected were mostly clay-sized samples, but some mudstones collected from the Idenohana Formation included silt-sized samples. The grain-size of the sandstone samples ranged from medium to fine sand.

In the present study, principal component analysis was conducted at first to discern the structure of the data. The MFW and $\text{SiO}_2/\text{Al}_2\text{O}_3$ – $\text{Na}_2\text{O}/\text{K}_2\text{O}$ diagrams were then utilized to measure the effects of weathering and hydraulic sorting processes.

CaO derived from carbonate was corrected using the method of McLennan (1993) and Bock *et al.* (1998). This manipulation first subtracts CaO in the phosphate phase using P_2O_5 concentration. If the molecular CaO is smaller than the molecular Na_2O , then CaO is accepted as approximate CaO^* (CaO in the silicate phase). If, however, CaO is greater than Na_2O , then molecular CaO^* is assumed to be equivalent to molecular Na_2O (McLennan, 1993). This manipulation was employed because it is difficult to estimate CO_2

Table 1. Major-element abundance in weight percent for the Ashikita Group.

	Sample no.	SiO ₂	TiO ₂	Al ₂ O ₃	Fe ₂ O ₃	MnO	MgO	CaO	Na ₂ O	K ₂ O	P ₂ O ₅	LOI	Total
Id sdst.	UM103103	51.47	0.57	12.72	5.39	0.11	2.49	13.31	3.11	1.11	0.12	9.08	99.48
	SA041705	64.82	0.79	14.36	6.52	0.11	2.67	1.69	5.24	0.97	0.10	1.61	98.88
	SA041702	62.37	0.84	15.12	7.16	0.12	2.81	1.91	4.99	0.84	0.11	1.03	97.30
	SA041606	51.02	0.72	12.93	6.25	0.12	2.67	12.62	3.16	0.99	0.15	9.41	100.04
	UM103102	55.20	0.67	13.42	6.54	0.09	3.01	9.00	3.13	1.33	0.14	6.25	98.78
	UM102704	59.05	0.68	14.13	6.33	0.09	2.78	5.90	3.76	1.33	0.14	5.23	99.42
	SA041303	64.82	0.69	15.81	5.23	0.08	2.15	1.90	5.19	1.33	0.06	1.08	98.34
Id mdst.	SA041404	56.18	0.84	16.98	7.30	0.09	3.27	5.38	2.91	2.28	0.19	4.24	99.66
	SA041402	56.11	0.77	16.43	6.91	0.08	3.18	6.23	2.84	2.21	0.18	4.37	99.31
	SA041403	54.88	0.75	15.56	6.89	0.09	2.96	7.50	3.06	1.93	0.18	6.31	100.11
	SA060305	63.81	0.66	20.27	5.74	0.01	1.98	0.46	1.96	3.93	0.11	0.91	99.84
	SA060308	66.66	0.71	17.69	5.70	0.04	1.87	0.35	2.24	3.60	0.11	1.23	100.20
	SA060301	65.59	0.82	17.31	5.15	0.06	2.21	1.41	2.96	3.14	0.17	1.20	100.02
	SA060303	64.80	0.72	18.60	6.12	0.05	2.01	0.49	2.18	3.64	0.16	1.13	99.90
Ky sdst.	SA031601	77.42	0.55	11.02	2.41	0.03	0.79	0.62	3.45	1.88	0.04	1.03	99.24
	SA031702	66.54	0.45	12.69	3.22	0.04	1.28	5.08	3.10	2.30	0.11	4.10	98.91
	SA041302	71.27	0.31	14.73	2.35	0.03	0.97	0.92	5.92	1.35	0.04	2.02	99.91
	SA041304	72.58	0.43	13.56	3.25	0.03	1.01	0.35	4.85	1.31	0.08	1.16	98.61
	SA031606	68.85	0.49	14.94	3.93	0.06	1.10	0.66	4.61	2.19	0.06	2.88	99.77
	UM102703	69.48	0.44	14.35	3.76	0.04	1.13	1.29	4.11	1.95	0.09	1.96	98.60
	SA032003	76.21	0.41	11.45	3.42	0.02	0.77	0.28	3.26	1.70	0.09	0.84	98.45
Ky mdst.	SA031610	72.03	0.48	13.55	3.19	0.02	0.86	0.31	5.04	1.54	0.07	1.22	98.31
	SA031617	70.58	0.50	14.79	2.62	0.03	1.00	0.38	4.80	2.28	0.06	1.78	98.82
	SA060102	63.40	0.70	17.60	7.43	0.05	1.95	0.85	2.50	3.80	0.15	1.98	100.41
	SA031605	66.72	0.68	17.27	5.20	0.06	1.94	0.72	2.48	3.35	0.13	1.13	99.68
	SA053101	62.95	0.73	19.25	6.39	0.03	1.98	0.42	2.21	4.48	0.14	1.96	100.54
	SA060103	64.05	0.71	18.13	6.08	0.05	1.97	0.80	2.57	3.83	0.17	1.73	100.09
	SA031604	66.61	0.70	17.72	5.69	0.03	1.57	0.68	2.13	3.47	0.16	1.47	100.23
Sk sdst.	SA031619	63.93	0.77	19.26	4.48	0.03	2.20	0.54	2.21	4.84	0.11	1.67	100.04
	SA060318	65.72	0.72	17.46	6.16	0.05	1.91	0.44	2.20	3.88	0.11	1.09	99.74
	SA060215	68.96	0.58	14.47	6.48	0.09	1.90	0.88	2.13	2.85	0.11	1.50	99.95
	SA060317	68.37	0.64	16.02	5.70	0.05	1.80	0.50	1.94	3.65	0.10	1.24	100.01
	SA053103	64.97	0.67	16.76	7.08	0.06	2.17	0.81	2.56	3.33	0.12	2.03	100.56
	SA060311	63.36	0.73	19.39	6.30	0.04	2.27	0.51	2.33	3.85	0.12	1.99	100.89
	SA041503	86.02	0.06	8.34	0.37	0.01	0.22	0.15	1.80	1.98	0.01	1.02	99.98
Sk mdst	SA031705	68.58	0.51	14.81	4.62	0.07	1.83	0.70	4.97	1.97	0.09	1.92	100.07
	SA031707	87.55	0.04	6.77	0.38	0.01	0.21	0.12	1.55	1.40	0.01	1.19	99.23
	SA041401	89.42	0.10	5.73	0.51	0.01	0.17	0.11	1.71	0.97	0.01	1.23	99.97
	SA041101	90.25	0.03	5.79	0.29	0.01	0.16	0.11	1.51	0.75	0.01	1.44	100.35
	SA032004	88.75	0.08	6.18	0.56	0.01	0.11	0.14	1.24	2.08	0.01	1.19	100.35
	SA031904	75.20	0.30	13.07	1.53	0.01	0.54	0.39	3.51	3.84	0.06	2.11	100.56
	SA041501	77.29	0.31	11.67	2.49	0.03	0.47	0.23	2.82	3.03	0.02	1.31	99.67
Sk mdst	SA041102	69.53	0.48	14.80	3.44	0.05	0.97	0.40	4.57	2.23	0.33	1.11	97.91
	UM102602	76.36	0.32	12.08	2.01	0.01	0.50	0.33	3.34	3.31	0.07	1.31	99.64
	SA031607	84.01	0.11	9.34	0.59	0.01	0.29	0.18	1.52	2.95	0.01	1.77	100.78
	SA060205	62.93	0.76	21.45	5.64	0.02	1.94	0.31	1.29	4.19	0.09	1.54	100.16
	SA060206	68.22	0.61	18.09	5.00	0.02	1.79	0.63	1.04	3.77	0.10	1.43	100.70
	SA060208	68.21	0.66	14.64	4.69	0.10	1.39	1.81	3.16	1.34	0.07	1.07	97.14
	SA060209	80.47	0.35	10.20	3.52	0.06	1.37	0.38	1.07	2.24	0.07	1.26	100.99
Sk mdst	SA060107	67.39	0.68	16.16	6.11	0.05	1.96	0.57	2.17	3.41	0.12	1.78	100.40
	SA041502	76.25	0.33	15.33	1.94	0.02	1.30	0.11	0.40	3.72	0.03	1.42	100.85
	SA060202	63.58	0.66	20.13	5.75	0.01	1.97	0.48	2.05	3.94	0.11	1.53	100.21
	SA060201	68.86	0.51	18.10	4.68	0.01	1.54	0.24	1.23	3.60	0.06	1.15	99.98
	SA060105	56.58	0.98	23.07	6.04	0.04	2.56	0.52	2.16	5.88	0.20	1.60	99.63

Id, Idenohana Formation; Ky, Kyodomari Formation; Sk, Sakamoto Formation; sdst., sandstone; mdst., mudstone.
[LOI = loss on ignition]

precisely from LOI (Table 1), which normally includes structural water, organic carbon, salt and carbonate carbon (e.g. Heiri *et al.*, 2001).

Geochemical data represented in compositions are subject to the constant-sum constraint. In this article, therefore, the centred logratio transformed data (Aitchison, 1986) were used as input data for principal component analysis. In addition, as diagrams in a Cartesian coordinate system were employed, compositional data were examined in the form of ratio or logratio (Aitchison, 1982, 1986). Linear regression lines within ternary diagrams were calculated using the statistical analytical technique designed for compositional data (see Aitchison, 1986; Aitchison & Egozcue, 2005; Ohta & Arai, 2006). All calculations were conducted by the statistical computing environment 'R' and additional package 'compositions' (van den Boogaart & Tolosana-Delgado, 2006).

RESULTS

Mineralogy of samples

For petrological description, modal compositions of 47 sandstone samples were evaluated. Modal counts of 300 points were performed for each sample following the Gazzi-Dickinson method (see Weltje, 2002). Samples can be classified into three groups according to their stratigraphic positions.

Sandstones of the Idenohana Formation are mostly classified into feldspathic wacke (Fig. 3A). Feldspar grains are predominantly composed of plagioclase, whereas K-feldspar is rarely present. In general, sandstones of the Idenohana Formation are significantly rich in intermediate to mafic volcanic rock fragments (Fig. 4A). One outlier sample plotted near the L vertex of the QFL

ternary diagram was extraordinarily rich in intermediate volcanic rock fragments (Fig. 3A). The composition of this anomalous sample is very informative for estimating the source rock lithology, but it was excluded from the following analysis because outliers commonly distort statistical results.

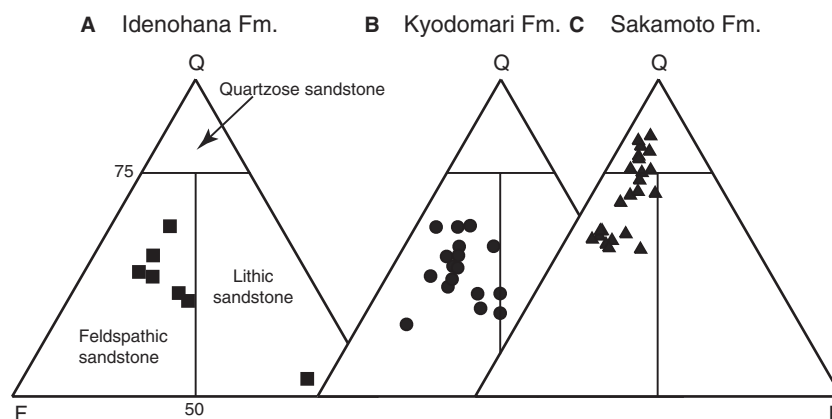
Sandstones of the Kyodomari Formation are poorly sorted and occasionally they contain grains larger than sand size. As in the underlying Idenohana Formation, the framework modes of sandstones are mostly feldspathic wacke (Fig. 3). However, the compositions of the rock fragments differ from those in the Idenohana Formation in that intermediate to mafic volcanic rock fragments decrease remarkably in this horizon. In contrast, the proportions of felsic volcanic rock fragments and granitic rock fragments increase in the Kyodomari Formation (Fig. 4B).

Within the Sakamoto Formation, quartz becomes the major constituent (60 to 80%) of the framework modes (Fig. 3C) and the abundance of volcanic rock fragments decreases abruptly in this horizon. In addition, the abundance of K-feldspar increases significantly compared with the underlying formations. The proportion of the K-feldspar in feldspar grains is about 15% in the Kyodomari Formation, whereas it increases to about 30% in the Sakamoto Formation. As a whole, the Sakamoto Formation shows a quartzose nature and the mineralogical maturity increases considerably (Fig. 4C).

Geochemistry of samples

Results of the geochemical analysis are listed in Tables 1 and 2. The principal component calculation was performed by the singular value decomposition of the centred logratio transformed data. The extracted principal components

Fig. 3. Modal composition of sandstones for the: (A) Idenohana, (B) Kyodomari and (C) Sakamoto Formations described in QFL diagram (Okada, 1971). Q, quartz; F, feldspar; L, lithic rock fragments.



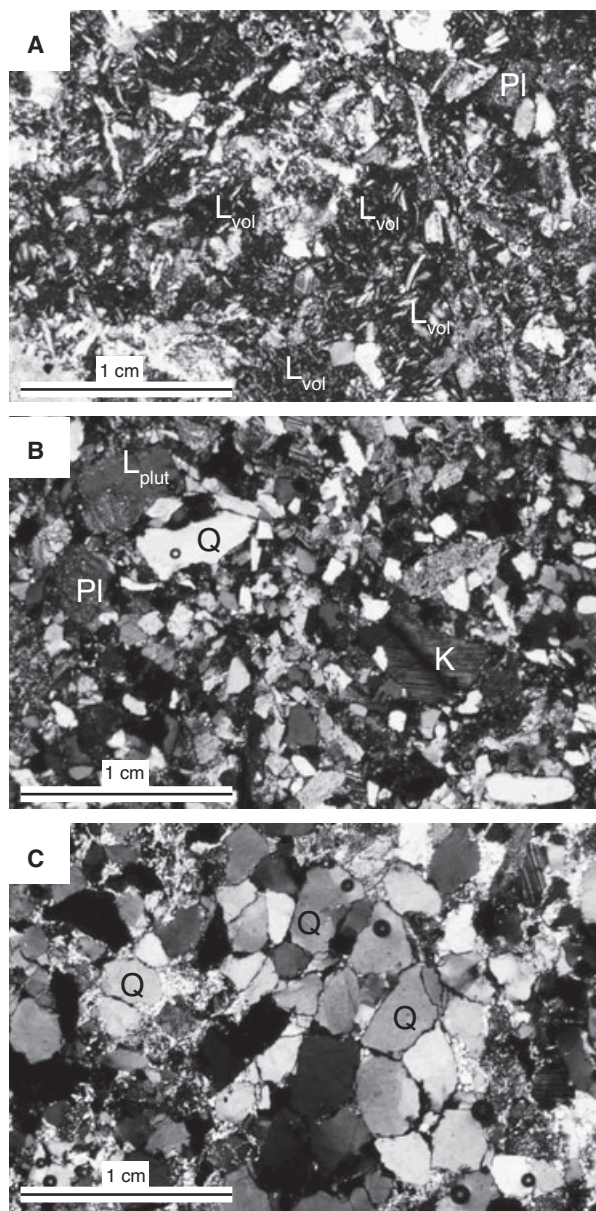


Fig. 4. Microphotographs of thin sections. (A) Lithic arenite from the Idenohana Formation containing abundant intermediate-mafic volcanic rock fragments (with cross-polarizers). (B) Ill-sorted feldspathic wacke from the Kyodomari Formation with abundant granitic rock fragments (with cross-polarizers). (C) Quartzose arenite from the Sakamoto Formation (with cross-polarizers). Q, quartz; Pl, plagioclase; K, potassic feldspar; L_{vol} , volcanic rock fragment; L_{plut} , plutonic rock fragment (granite).

can be seen as latent variables that best explain the variance of multivariate data. Table 3 gives the result of the first four principal components and Fig. 5 illustrates the result graphically. Collectively, the first two principal components accounted for 65.48% of the total variance of the geochemical data (Table 3).

Principal component 1 (PC1) shows positive loadings with immobile elements (e.g. SiO_2 , Al_2O_3 , Zr and Th), suggesting positive correlations between PC1 and these elements (Table 3). Generally, these elements are enriched in felsic igneous rocks. On the other hand, the highest negative loadings are found in ferromagnesian elements (e.g. Fe_2O_3 , MgO, Co and V). Ferromagnesian elements are enriched preferentially in mafic igneous rocks although they may show variation because of weathering (Feng & Kerrich, 1990; Garver *et al.*, 1996; Armstrong-Altrin *et al.*, 2004). Sandstones of the Idenohana Formation, which are rich in intermediate to mafic volcanic rock fragments, also display negative scores on the biplot (Fig. 5). Therefore, PC1 seems to represent a latent variable that projects a compositional variation due to the parent igneous rock types. Note that samples of each formation are best discriminated in the direction of the PC1 (Fig. 5). The increase in the PC1 scores from the Idenohana to the Sakamoto Formation indicates a systematic change from relatively mafic to more felsic compositions in ascending stratigraphic order. This result is consistent with the petrological observations of the sandstones described above.

Principal component 2 (PC2) displays the highest loadings with Na_2O , CaO and Sr, which are the main elements leached and carried away by groundwater during weathering processes (e.g. Nesbitt *et al.*, 1980; Railsback, 2003). In contrast, Al_2O_3 , Fe_2O_3 , TiO_2 , Nb, Th and Ni tend to show negative to nearly zero loadings (Table 3). During weathering processes, these elements tend to remain in soils (e.g. Nesbitt *et al.*, 1980; Railsback, 2003). The result of PC2 is consistent with the geochemistry of weathering processes observed in recent weathering profiles (e.g. Nesbitt & Markovics, 1997; White *et al.*, 2001; Duzgoren-Aydin *et al.*, 2002; Girty *et al.*, 2003; Turner *et al.*, 2003).

However, it should also be noted that PC2 clearly separates the sandstone and mudstone samples (Fig. 5). This relationship implies that fractionation of sand grains and phyllosilicates by hydraulic sorting processes is another important factor affecting PC2. In addition, PC2 shows a high positive loading with Zr (Table 3), indicating a concentration of resistant heavy minerals such as zircon in the sand-sized fraction. Therefore, PC2 is interpreted as a composite latent variable representing the effects of source-area weathering and hydraulic sorting processes.

The geochemical composition of the Ashikita Group is governed principally by the composition

Table 2. Minor-element abundance in p.p.m. for the Ashikita Group.

	Sample no.	Ba	Co	Cr	Cu	Nb	Ni	Pb	Rb	Sr	Th	V	Y	Zr
Id sdst.	UM103103	340	18	55	21	4	16	8	27	325	3	133	19	93
	SA041705	207	20	68	24	4	17	5	19	249	4	194	22	125
	SA041702	224	25	68	25	4	21	6	18	287	2	215	26	132
	SA041606	185	23	73	16	4	19	8	24	348	3	174	21	117
	UM103102	243	22	76	16	4	21	9	32	331	3	143	20	119
	UM102704	310	24	81	21	4	23	10	32	315	4	148	20	121
	SA041303	606	17	61	2	4	13	7	25	513	3	137	16	119
Id mdst.	SA041404	293	24	70	45	6	28	13	73	249	5	164	27	135
	SA041402	274	28	68	42	6	30	16	69	260	6	159	25	123
	SA041403	245	29	71	40	6	29	18	58	279	5	154	25	122
	SA060305	1095	14	47	9	16	25	32	189	147	20	111	24	134
	SA060308	429	16	64	36	13	29	24	169	91	14	171	28	157
	SA060301	366	19	68	23	8	26	15	118	217	8	130	24	176
	SA060303	459	20	59	33	13	23	24	156	104	13	119	28	162
Ky sdst.	SA031601	449	12	92	3	6	11	9	48	207	5	76	14	268
	SA031702	472	15	55	2	7	16	20	76	292	8	90	19	154
	SA041302	553	15	33	4	3	6	8	25	431	2	74	14	80
	SA041304	323	12	64	8	7	14	15	51	132	7	80	14	129
	SA031606	1113	15	71	3	7	16	12	64	253	7	103	33	136
	UM102703	308	17	43	2	6	16	10	59	210	5	86	19	119
	SA032003	440	12	38	3	10	7	12	69	90	10	55	24	186
Ky mdst.	SA031610	699	10	64	5	7	14	11	46	174	6	78	14	145
	SA031617	912	7	66	5	7	11	8	69	195	7	111	13	139
	SA060102	532	25	61	38	12	34	30	158	139	15	134	29	147
	SA031605	437	17	55	44	12	29	24	155	129	14	117	31	146
	SA053101	399	20	61	41	13	27	26	191	85	14	131	28	155
	SA060103	475	21	64	42	13	27	24	164	140	15	135	31	148
	SA031604	491	14	55	51	12	33	25	155	132	13	124	31	141
	SA031619	570	15	76	39	13	26	19	217	98	13	152	31	152
	SA060318	431	17	65	31	13	28	20	183	74	12	131	28	150
	SA060215	557	19	49	34	11	23	21	127	125	11	105	27	121
	SA060317	493	16	45	29	12	25	25	179	88	13	134	25	141
	SA053103	465	19	55	45	11	26	26	149	138	13	114	27	142
	SA060311	342	16	66	49	12	26	26	163	104	15	143	25	147
Sk sdst.	SA041503	322	1	20	0	3	4	13	74	72	3	7	7	69
	SA031705	679	17	55	3	7	17	14	57	196	8	96	21	160
	SA031707	228	2	16	0	2	5	12	56	50	3	8	8	48
	SA041401	160	2	15	0	3	3	9	42	42	5	9	8	82
	SA041101	885	12	66	3	7	15	13	62	205	8	91	19	140
	SA032004	419	4	12	1	3	2	15	74	56	4	4	8	97
	SA031904	724	5	37	2	7	5	17	127	169	9	45	14	133
	SA041501	835	7	36	0	8	9	16	95	140	14	46	22	164
	SA041102	81	1	8	1	2	7	10	32	32	3	8	8	42
	UM102602	659	6	42	3	8	8	15	111	132	11	47	15	145
	SA031607	565	0	16	0	3	3	16	103	68	3	14	8	64
Sk mdst	SA060205	267	18	58	13	14	33	28	191	90	15	116	36	110
	SA060206	376	19	67	18	17	30	30	205	97	21	130	28	136
	SA060208	769	12	37	18	11	14	19	110	89	10	71	48	91
	SA060209	464	17	55	13	17	23	23	225	77	19	127	27	131
	SA060107	414	21	60	39	12	34	24	151	109	13	128	25	131
	SA041502	507	5	19	10	13	9	8	169	28	27	41	39	124
	SA060202	809	14	50	10	16	24	30	190	147	20	111	23	133
	SA060201	359	18	47	14	22	21	34	188	85	20	101	39	123
	SA060105	481	22	86	33	20	29	39	264	87	22	139	39	144

Id, Idenohana Formation; Ky, Kyodomari Formation; Sk, Sakamoto Formation; sdst., sandstone; mdst., mudstone.

Table 3. PCA loadings for PC1 and PC2.

	PC1	PC2	PC3	PC4
SiO ₂	0.7972	0.4939	-0.1493	0.1475
TiO ₂	-0.6541	-0.1661	-0.5122	-0.1898
Al ₂ O ₃	0.6364	0.2529	-0.5703	-0.0301
Fe ₂ O ₃	-0.8072	-0.2714	-0.3701	-0.0648
MnO	-0.5601	0.4744	-0.1963	0.5058
MgO	-0.8502	-0.1939	-0.2689	0.0618
CaO*	-0.674	0.5614	-0.1450	0.1419
Na ₂ O	0.1334	0.8825	-0.2529	-0.1403
K ₂ O	0.8177	-0.2649	-0.3764	-0.1713
P ₂ O ₅	-0.6738	-0.2121	-0.5167	-0.2168
Ba	0.7949	0.3223	0.1839	-0.0589
Co	-0.6237	-0.1042	0.5602	0.0271
Cr	0.0095	0.5070	0.7119	-0.2818
Cu	-0.6082	-0.6089	0.1574	0.0664
Nb	0.7277	-0.5895	0.2083	-0.0411
Ni	-0.3926	-0.5197	0.3917	-0.0410
Pb	0.8345	-0.2843	-0.1330	-0.0756
Rb	0.8141	-0.5291	-0.0600	-0.0421
Sr	-0.1076	0.9101	0.2284	-0.1139
Th	0.7568	-0.5762	0.0685	-0.0154
V	-0.6709	-0.0956	0.6183	-0.1297
Y	0.5330	-0.2166	0.1915	0.6873
Zr	0.7266	0.4608	0.1993	-0.0920
Eigenvalues	10.0311	5.0304	2.9642	1.0485
Proportion	43.61	21.87	12.89	4.56
Cumulative proportion	43.61	65.48	78.37	82.93

PCA, principal component analysis; CaO*, CaO in silicate phase.

of the source rocks, as well as weathering and hydraulic sorting effects. Therefore, the present results verify the need to compensate for the effects of weathering and hydraulic sorting processes in order to achieve a robust provenance analysis, as suggested by many previous workers (Ingersoll *et al.*, 1984; Argast & Donnelly, 1987; Johnsson, 1993; Nesbitt & Young, 1996; Nesbitt *et al.*, 1996).

DISCUSSION

Consequences of source-area weathering

The result of principal component analysis indicates that source-area weathering and hydraulic sorting are extremely significant in modification of the original composition of the source rocks for the sediments of the Ashikita Group. Figure 6 illustrates the MFW diagram, which is developed from a statistical empirical index of chemical weathering (Ohta & Arai, 2007). The M and F vertices represent unweathered mafic and felsic

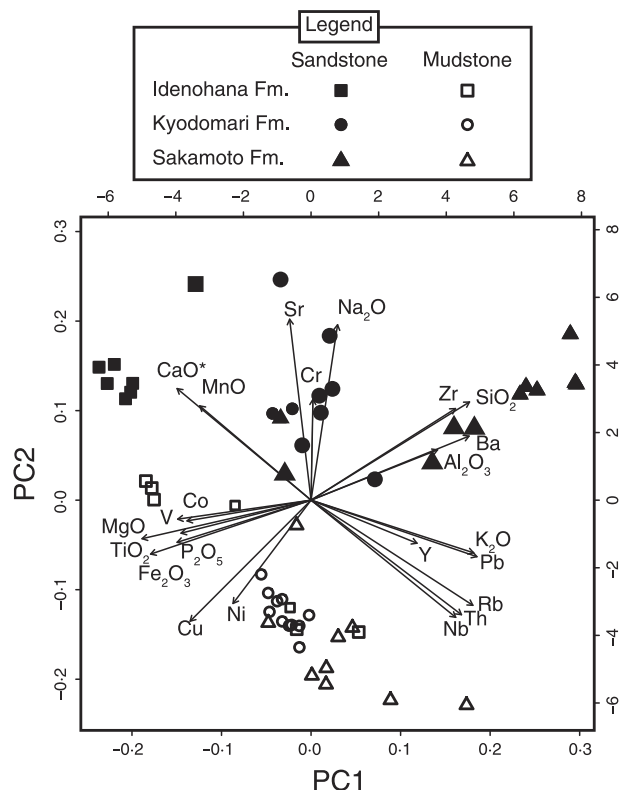


Fig. 5. Results of principal component analysis described as compositional biplot. Arrows indicate loadings of elements.

igneous rocks, respectively. The W vertex, on the other hand, measures the degree of weathering of these parent igneous rocks. For reference purposes, some standard igneous rock samples provided by the Geological Survey of Japan (Imai *et al.*, 1995) are plotted in Fig. 6D.

Weathering trends on the MFW diagram are characterized primarily by the loss of Na₂O and CaO and relative enrichment of Fe₂O₃, Al₂O₃ and K₂O during weathering (Ohta & Arai, 2007). This elemental behaviour is generally concordant with PC2 (Fig. 5), which is interpreted as the weathering trend of the sediments in the Ashikita Group. Accordingly, the MFW diagram seems to be suitable for predicting the weathering trends of sedimentary rocks of the Ashikita Group.

On the MFW diagram, mudstones that are composed of phyllosilicates depict high W values (Fig. 6). However, sandstones that are composed chiefly of weathered residues of the source rock depict low W values in comparison with the mudstones. As a result, sandstones and mudstones delineate a compositional linear trend (Aitchison, 1983, 1986; von Eynatten *et al.*, 2003a) on the MFW diagram extending to the W vertex (Fig. 6). Therefore, the compositions of the

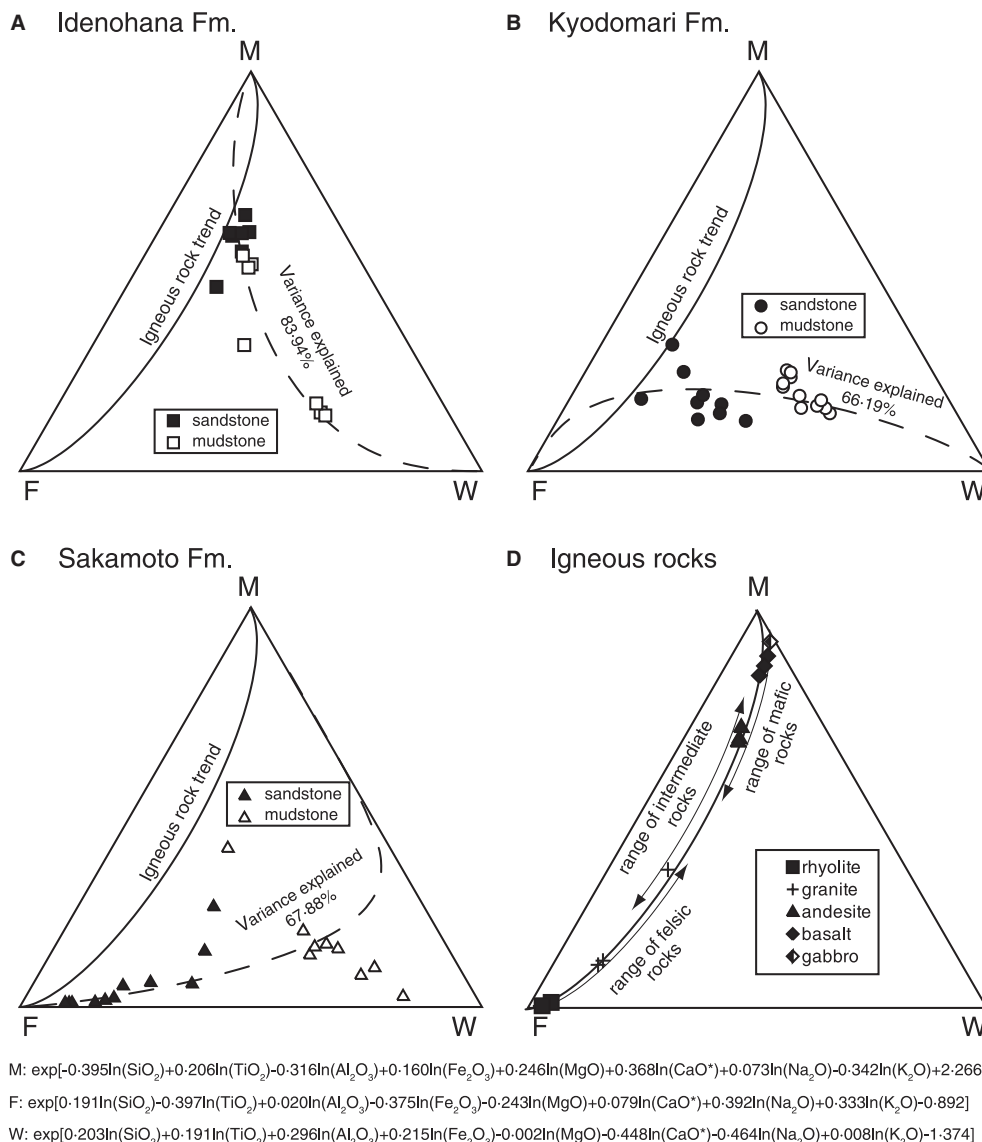


Fig. 6. Weathering trends depicted on the MFW diagram (Ohta & Arai, 2007) for the: (A) Idenohana Formation, (B) Kyodomari Formation, (C) Sakamoto Formation and (D) plots for some common igneous rocks (Imai *et al.*, 1995). Weathering trends of the Idenohana and Kyodomari Formations intersect with the igneous rock trend in the plot domain of andesites and granites, respectively. For the sediments of the Sakamoto Formation, the weathering trend does not extend towards the W vertex, indicating that they cannot be represented by the weathering of the igneous rocks.

sediments reflect the compositions of weathering or sorting profiles of the source rock, rather than the composition of the source rock itself. It is difficult to obtain direct source composition information from raw geochemical data. However, on the MFW diagram, the effect of weathering can be adjusted by tracing back these trends to the domain of unweathered igneous rocks.

The weathering trends of the Idenohana and Kyodomari Formations extend backwards to the M and F vertices, respectively (Fig. 6A and B). The weathering trend of the Idenohana Formation

intersects with the igneous rock trend where unweathered andesites are plotted (Fig. 6A and D). Similarly, the weathering trend of the Kyodomari Formation passes through the domain of granites (Fig. 6B and D). Consequently, it is inferred that the source rocks of the Idenohana and Kyodomari Formations had an average composition comparable to that of intermediate and felsic igneous rocks, respectively. This interpretation is consistent with the petrographical observations of the sandstones (Figs 3 and 4). The rock fragments of the Idenohana and Kyodomari

Formations were mostly intermediate and felsic volcanic rock fragments, respectively.

The weathering trend of the Sakamoto Formation intersects with the igneous rock trend in the proximity of the F vertex (Fig. 6C), indicating that the source composition was more felsic compared to that of the other two formations. However, the weathering trend of the Sakamoto Formation does not extend towards the W vertex, implying that the MFW diagram is ineffective in describing the weathering trend in this case (at least in linear regression). As mentioned earlier, the MFW diagram is based on data of weathering profiles developed on igneous rocks; this feature indicates that the weathering trend of the Sakamoto Formation cannot be approximated by the weathering trends of igneous rocks. It is therefore necessary to consider source rock candidates other than igneous rocks for the sediments of the Sakamoto Formation.

In summary, the MFW diagram facilitates provenance analysis of sedimentary rocks by identifying their weathering trends and thereby enabling a backward estimate of the average geochemical composition of the unweathered source rock. The backward estimate method clearly determined the differences in the source area composition of the Ashikita Group. However, the main limitation of the MFW diagram is revealed when source-area composition cannot be approximated to the composition of igneous rocks.

Consequences of the hydraulic sorting effect

In addition to weathering, the effect of hydraulic sorting and grain-size fractionation results in considerable modification of sediment compositions (Ingersoll *et al.*, 1984; Argast & Donnelly, 1987; Garcia *et al.*, 1991; Nesbitt & Young, 1996; Ohta, 2004). Moreover, Ethridge (1977) and LaFlèche & Camire (1996) showed that transportation and sorting processes might produce mineralogically mature sediments.

Hydraulic sorting processes lead to contrasting distribution patterns of minerals in different size fractions (clay to very coarse sand), including differential concentration of heavy minerals. Ohta (2004) demonstrated that $\text{SiO}_2/\text{Al}_2\text{O}_3$ and $\text{Na}_2\text{O}/\text{K}_2\text{O}$ ratios are highly susceptible to the fractionation of quartz, albite and illite during sorting effects. Therefore, these ratios can be utilized to measure a simple binary response of phyllosilicate and sand-sized particle end-members during hydraulic sorting processes.

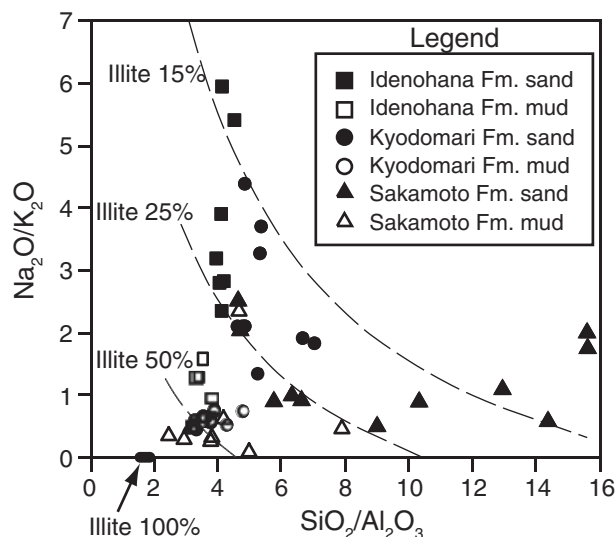


Fig. 7. Plots of the $\text{SiO}_2/\text{Al}_2\text{O}_3$ – $\text{Na}_2\text{O}/\text{K}_2\text{O}$ diagram designed to measure the effect of hydraulic sorting (Ohta, 2004). Sediments delineate consecutive arrays depending on the extent of winnowing of phyllosilicates. Dashed lines are calculated contours for 50%, 25% and 15% matrix compositions for a hypothetical sediment composed of quartz, albite and illite (for a detailed explanation, see Ohta, 2004). On this diagram, sediments derived from an igneous provenance delineate vertical trends as shown by those of the Idenohana and Kyodomari Formations. In contrast, sediments derived from a recycled sedimentary provenance delineate horizontal trends as shown by those of the Sakamoto Formation.

On the $\text{SiO}_2/\text{Al}_2\text{O}_3$ – $\text{Na}_2\text{O}/\text{K}_2\text{O}$ diagram, mudstones plot near the compositional domain of illite (Fig. 7). In contrast, sandstones which are deficient in matrix-sized illite plot progressively at a distance from the illitic composition, which should be reflecting the sorting and winnowing effect on phyllosilicates (Fig. 7). Consequently, sandstones and mudstones delineate compositional trends, in which the geochemical variability induced by hydraulic sorting is expressed by the range and extent of these trends. The source rock composition, prior to modification by the sorting effect, can be quantified simply by tracing back the trend to a point of no sorting (Ohta, 2004).

In the present example, sediments of the Ashikita Group delineate two nearly orthogonal trends (Fig. 7). The compositional dispersion resulting from the hydraulic sorting effect should be restricted within the range of these trends. Therefore, the hydraulic sorting process cannot be extended to account for the differences in the slope angles. According to Ohta (2004), slopes in the $\text{SiO}_2/\text{Al}_2\text{O}_3$ – $\text{Na}_2\text{O}/\text{K}_2\text{O}$ diagram are

determined by the initial source composition. Sediments exclusively derived from igneous provenances will be arranged vertically on the diagram. Sediments derived from a quartz-rich recycled sedimentary provenance will be arranged horizontally (Ohta, 2004). The relationship in Fig. 7, therefore, indicates that sediments of the Idenohana and Kyodomari Formations were sourced primarily from igneous rocks and those of the Sakamoto Formation were sourced from recycled sedimentary rocks.

Consequences of diagenesis

Diagenetic processes are known to modify the geochemical composition of sedimentary rocks (e.g. Fedo *et al.*, 1995). Petrographical observations suggest that cementation by silica is uncommon for the samples of the Ashikita Group. However, as can be seen in Fig. 4A, cementation by carbonate is common especially for several samples in the Idenohana Formation. This effect results in geochemical modification that apparently involved addition of CaO and Sr in some samples. To minimize the effect of carbonate cementation, CaO* (CaO in the silicate phase) has been used consistently in this article.

In ancient sedimentary sequences, the addition of K₂O by K-metasomatism is common (Fedo *et al.*, 1995; Hemming *et al.*, 1995; Bock *et al.*, 1998). Nesbitt & Young (1989) and Fedo *et al.* (1995) demonstrated a method for evaluating K₂O addition using the A-CN-K diagram (Fig. 8). In this diagram, unmetasomatized sandstones and mudstones will delineate an ideal weathering trend, which extends nearly parallel to the A-CN axis. If there were an addition of K₂O by diagenesis, the weathering trend would be arranged oblique to the A-CN axis (Fedo *et al.*, 1995).

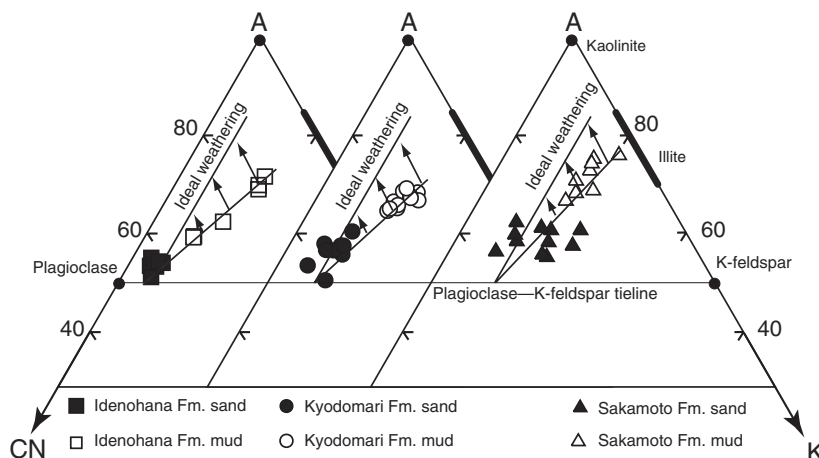
Sediments of the Idenohana, Kyodomari and Sakamoto Formations are arranged oblique to the ideal weathering trend (Fig. 8). These inclinations probably indicate K₂O addition during diagenesis. Using this deviation from the ideal weathering trend, possible premetasomatized compositions of sediments were calculated by subtracting the excess K₂O added by metasomatism. The MFW and SiO₂/Al₂O₃–Na₂O/K₂O diagrams for recalculated premetasomatized compositions are shown in Fig. 9. Although there are some differences in uncorrected raw plots (Figs 6 and 7) and corrected premetasomatized plots (Fig. 9A and B), the general trends that each formation depicted are closely analogous. Together, these lines of evidence demonstrate that the contribution from diagenesis is insignificant as long as CaO in the carbonate phase is corrected.

Provenance

In the discussion above, both the MFW and SiO₂/Al₂O₃–Na₂O/K₂O diagrams revealed compositional variations among the three formations, even after adjusting for the effects of weathering and hydraulic sorting processes (Figs 6 and 7). Therefore, the source rocks for each formation must be different in order to create distinctive compositional trends on both diagrams. This interpretation is further reinforced by a ternary diagram using SiO₂–Na₂O–CaO* (Fig. 10), which is designed to distinguish igneous and recycled sedimentary provenances.

On the biplot (Fig. 5), sandstones of the Idenohana, Kyodomari and Sakamoto Formations are generally dispersed along the vectors of CaO*, Na₂O and SiO₂, respectively. Therefore, these three components best describe the

Fig. 8. A-CN-K diagrams and corresponding CIA values. Note that weathering trends for the Idenohana, Kyodomari and Sakamoto Formations deviate from the ideal weathering trend, indicating possible K-metasomatism during diagenetic processes.



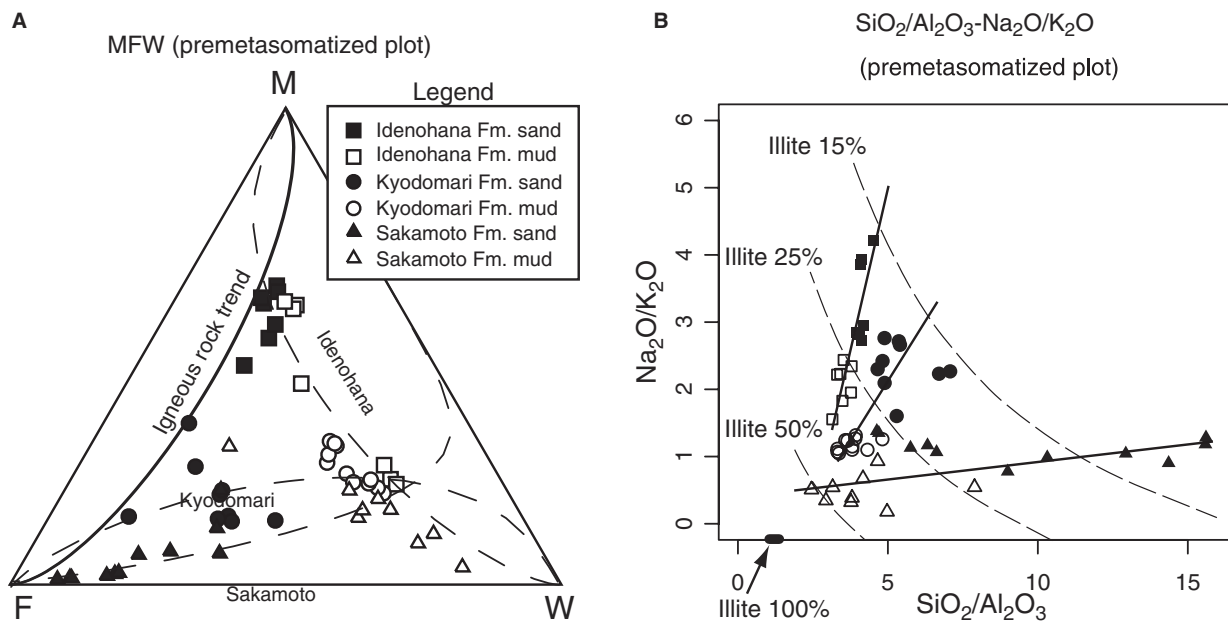


Fig. 9. Plots of recalculated premetasomatized sandstone and mudstone compositions on the MFW (A) and $\text{SiO}_2/\text{Al}_2\text{O}_3\text{--Na}_2\text{O}/\text{K}_2\text{O}$ (B) diagrams.

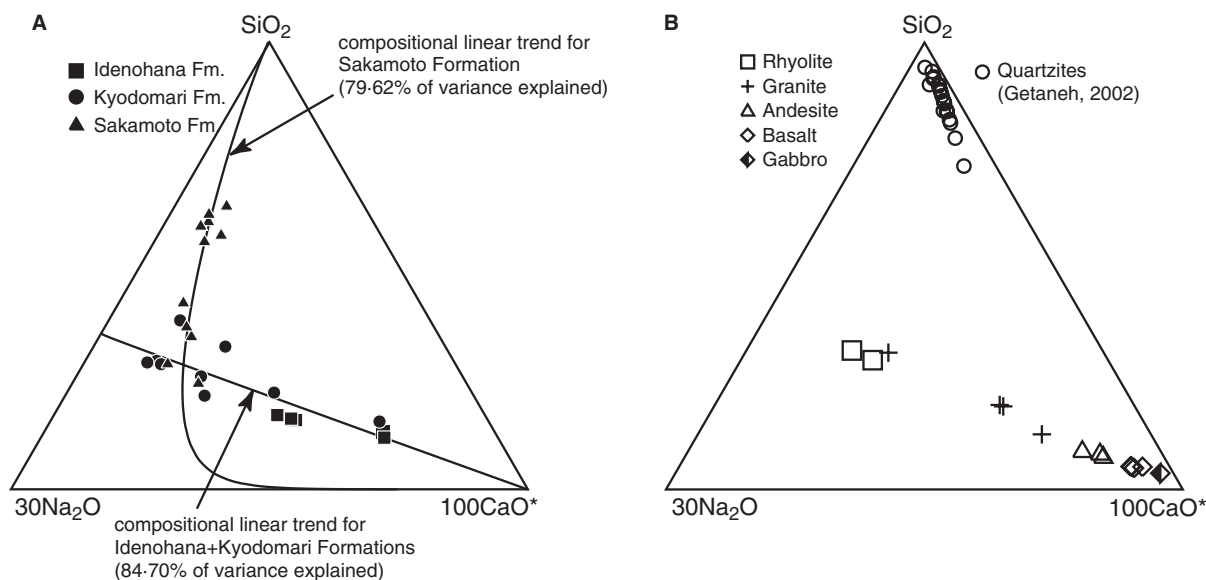


Fig. 10. Ternary diagrams of $\text{SiO}_2\text{--Na}_2\text{O--CaO}^*$ for: (A) the sandstones of the Ashikita Group; and (B) reference plots for common igneous rocks (Imai *et al.*, 1995) and quartzites (Getaneh, 2002). Na_2O and CaO^* components were multiplied by 30 and 100, respectively, for centring the plots. The plot regions for the Idenohana and Kyodomari Formations resemble those of the igneous rocks. Plots of the Sakamoto Formation connect the plot domains of felsic igneous rocks and quartzites, implying a mixed source from these two lithologies.

compositional variation of the sandstones. On Fig. 10B, some typical igneous rocks and quartzites are plotted for reference purposes. As is evident from Fig. 10B, this ternary diagram provides a criterion for distinguishing between igneous provenance and recycled sedimentary provenance.

Sandstones of the Idenohana and Kyodomari Formations are arranged in a straight line, repre-

senting constant SiO_2 and Na_2O fractions (Fig. 10A). The slope of the straight line and the range of sandstone distribution are fully concordant with the plots of the igneous rocks, especially with those of felsic to intermediate rocks. However, sandstones of the Sakamoto Formation depart from this straight line towards the SiO_2 vertex. The compositional linear trend of the Sakamoto Formation connects the plot domain of felsic igneous

rocks and those of quartzites, which seems to display a mixing line between these two end-members (Fig. 10A). Using these geochemical and petrographical results, the provenances of each formation can be interpreted as follows.

Figures 6, 7 and 10 suggest that sediments of the Idenohana Formation were supplied from an igneous provenance. Lithic fragments of the Idenohana Formation are predominantly intermediate volcanic rock fragments and, subordinately, mafic volcanic rock fragments. This observation implies that the sediments were derived mainly from an intermediate volcanic rock source and subordinately from a mafic volcanic rock source. In addition, as sandstones universally contain quartz grains, felsic igneous rocks must have contributed some of the sediment. Consequently, the mixture of these lithologies resulted in a bulk geochemical composition comparable to that of intermediate igneous rocks (Fig. 10A).

The MFW and $\text{SiO}_2\text{-Na}_2\text{O-CaO}^*$ diagrams (Figs 6 and 10) indicate that the sediments of the Kyodomari Formation are felsic in composition relative to those of the Idenohana Formation. This result is supported by the framework modes of sandstones, which were rich in K-feldspar, felsic volcanic rock fragments and granitic rock fragments. The ratio of Plagioclase to K-feldspar was about 85:15, which is equivalent to the modal composition of granodiorite. Therefore, felsic igneous rocks are inferred as the chief source rock of the Kyodomari Formation.

In Figs 7 and 10A, plots of the Sakamoto Formation have geochemical characteristics similar to those of the quartzites. In addition, the petrological features of the Sakamoto Formation denote a high maturity. Such mature sediments are rare in an active continental arc setting and sediment supply from a recycled sedimentary provenance must have been involved in the generation of orthoquartzitic sediments. However, extreme source-area weathering could also produce first-cycle mature sediments. The W and CIA values for the sandstones and mudstones of the Sakamoto Formation have ranges similar to those of the Idenohana and Kyodomari Formations (Figs 6 and 8). Therefore, it is apparent that the sediments of the Ashikita Group experienced similar weathering conditions and it is unreasonable to assume that only the sediments of the Sakamoto Formation underwent severe source rock weathering; it is more likely that they had a recycled sedimentary provenance.

The siliciclastic successions of the Kurosegawa Belt, including the Ashikita Group, are regarded

as a forearc basin infill of the eastern Asian continental arc (e.g. Sakai & Okada, 1997). Judging from the southerly-directed palaeocurrent (Fig. 2), the igneous rock provenances of the Ashikita Group possibly belong to the magmatic arc of the eastern Asian continental margin. The recycled sedimentary provenance, which was the source for sediments of the Sakamoto Formation, probably belongs to the continental interior region of the Asian continent.

CONCLUSIONS

Practical application of the MFW and $\text{SiO}_2/\text{Al}_2\text{O}_3\text{-Na}_2\text{O/K}_2\text{O}$ diagrams to the Ashikita Group enabled measurement and adjustment of weathering and hydraulic sorting effects. In addition, the diagrams unravelled the source-area lithology of the sediments. Sediments of the Idenohana and Kyodomari Formations were supplied from an igneous rock provenance. The source rocks of the Idenohana and Kyodomari Formations had changed from a composition comparable to that of intermediate igneous rock to felsic igneous rock. In contrast, the source rock of the Late Jurassic Sakamoto Formation is composed chiefly of recycled sedimentary rocks.

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